

TITLE: AN OCR-BASED REAL ESTATE MANAGEMENT SYSTEM FOR ST. FRANCIS SQUARE REALTY CORPORATION USING LEVENSHTTEIN DISTANCE FOR AUTOMATED DOCUMENT VALIDATION

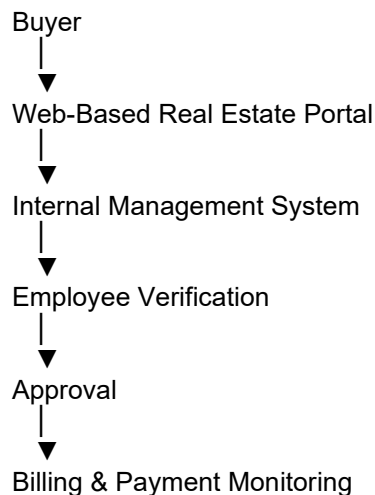
Project Overview:

The proposed system is composed of **two integrated platforms**:

1. **Web-Based Real Estate Portal** – used by prospective buyers and existing clients.
2. **Internal Management System** – used by authorized employees to manage reservations, documents, billing, and reports.

Both platforms share a **single centralized database**, meaning any update made in one system is immediately reflected in the other.

Sample Workflow



1. Reservation Process Through the Web-Based Portal (Online Buyer)

Step 1 – Browse Available Units

The buyer visits the website and browses available condominium units without creating an account.

They can view:

- Unit details
- Floor plans
- Amenities
- Prices
- Promotions

Step 2 – Create an Account

If interested, the buyer registers an account.

After registration, the buyer can:

- Schedule site tripping
- Reserve a unit
- Upload documents
- Monitor reservation status

Step 3 – Reserve a Unit

The buyer selects an available unit and submits a reservation request.

The system immediately places the selected unit under **"On Hold"** status so that another buyer cannot reserve the same unit while it is being processed.

Step 4 – Upload Documentary Requirements

The buyer uploads all required documents, such as:

- Valid IDs
- Proof of Billing
- Income Documents
- Reservation Forms
- Other required requirements
- Proof of Reservation Payment

These files are automatically stored in the centralized database.

2. Reservation Process Through the Internal Management System (Walk-in Buyer)

Not all buyers reserve online. Some buyers visit the office personally.

In this situation, employees will create the reservation using the Internal Management System.

Step 1 – Create Reservation

The Sales or Documentation staff manually enters:

- Buyer's information
 - Selected unit
 - Reservation date
 - Contact information
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Step 2 – Unit Status Changes

After saving the reservation, the system automatically changes the unit status to **"On Hold"** to prevent double selling.

Step 3 – Upload Buyer Documents

If the buyer submits printed or scanned documents, the employee uploads them into the system.

Examples include:

- Government IDs
- Reservation Forms
- Proof of Billing
- Income Documents
- Reservation Payment Receipt

The uploaded documents are linked to the buyer's record.

Step 4 – OCR Document Validation

After uploading, the system automatically starts the document validation process.

OCR extracts important information such as:

- Full Name
 - Document Type
 - ID Number
 - Date
 - Address
-

Step 5 – Levenshtein Distance Validation

The extracted text is compared with the buyer's registration details.

If OCR reads:

JUAN DELA CRVZ

instead of

JUAN DELA CRUZ

The Levenshtein Distance Algorithm recognizes that only one character is different and considers it a valid match, reducing unnecessary document rejection.

Step 6 – Employee Review

The employee reviews:

- OCR results
- Validation results
- Completeness of requirements

The employee can:

- Approve
 - Reject
 - Request additional documents
-

Step 7 – Reservation Approval

Once everything is complete:

The reservation is approved.

The system automatically changes the unit status from:

On Hold

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Reserved / Sold

The unit is immediately removed from the list of available units on the Web Portal.

3. Document Upload Process

Documents can be uploaded in **two ways**:

A. Buyer Uploads Documents (Web Portal)

Buyer logs into the Client Portal.

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Selects the required document.

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Uploads the file.

↓

System saves the file.

↓

OCR starts automatically.

↓

Employee reviews the results.

B. Employee Uploads Documents (Internal Management System)

Walk-in buyer submits physical documents.

↓

Employee scans or uploads the documents.

↓

Files are attached to the buyer's record.

↓

OCR starts automatically.

↓

Employee reviews the validation.

4. Document Validation Process

Once a document is uploaded, the system performs these steps automatically:

Step 1

OCR reads the uploaded document.

↓

Step 2

Important information is extracted.

↓

Step 3

Levenshtein Distance compares the extracted information with the buyer's registered information.

↓

Step 4

The system determines whether the document matches.

↓

Step 5

Employee reviews the validation results before making the final decision.

5. Billing Process

Once the reservation is approved:

The buyer can:

- View the Statement of Account (SOA)
- Upload proof of payment
- Monitor balances
- View payment history

Employees verify uploaded payments before posting them.

6. Cancellation Process

If documentary requirements are not completed within **30 days**:

The reservation is manually cancelled by the employee, subject to management approval.

The system then:

- Changes the reservation status to **Cancelled**
- Keeps all records for future reference
- Returns the unit to **Available** status

ONE SAMPLE OF SYSTEM WORK FLOW

ONLINE BUYER

Browse Units



Register Account



Reserve Unit



System tags Unit as ON HOLD



Upload Required Documents



OCR Reads Documents



Levenshtein Distance Validates Information

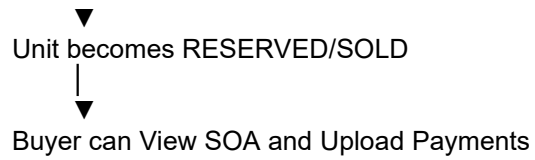


Employee Reviews Results



Approve Reservation





WALK-IN BUYER

